

Atomic Resolution Strain Imaging of nano scale materials with Bragg Coherent Diffraction Imaging

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Abstract

Bragg coherent diffractive imaging (BCDI) can resolve the three-dimensional strain field of a micron-scale crystal, but standard iterative phase-retrieval algorithms recover only a smoothed electron density rather than atomic positions. PRAMMol is an alternative that embeds molecular dynamics directly in the reconstruction loop, treating every atom as a free parameter constrained to physically realistic configurations. This report describes a from-scratch, GPU-accelerated reimplementation of PRAMMol that replaces four of its most expensive stages — the log-likelihood gradient, the structure-factor evaluation, candidate relaxation, and candidate refinement — with custom GPU kernels and a non-uniform FFT, making reconstructions at the 5k–50k-atom scale computationally tractable. Applied to simulated tungsten grains exposed to radiation damage, convergence is characterized across a 144-run library and found to depend strongly on grain size (70.5% convergence at 5k atoms, falling to 0% at 50k) and, counterintuitively, to improve with radiation dose rather than degrade: an undamaged control fails to converge where every damaged grain in the same series converges cleanly. These results establish PRAMMol’s potential to image the strain state of irradiated nanoscale materials directly.

1 Introduction

Strain at the atomic scale governs how materials behave and ultimately fail: it drives dislocation motion and fracture in structural metals, degrades capacity in battery electrodes over repeated cycling, and shifts the electronic and catalytic properties of metal nanoparticles. Understanding and imaging this strain is therefore valuable across a wide range of technologies, from mechanical design to energy storage and catalysis. Because the strain fields are often produced by individual crystal defects, imaging them faithfully requires resolving atomic positions directly, a scale that most conventional imaging techniques cannot reach with both precision and depth.

Bragg coherent diffractive imaging (BCDI) is a technique with the potential to achieve atomic resolution of micron-sized crystals [1]. In BCDI, X-rays are scattered from the crystal and measured in a region around a Bragg peak, as shown in Figure 1. During measurement, only the intensity information is gathered; the photon phase data is lost. While we cannot directly measure the phase data, the needed information is encoded in the diffraction pattern. We can reconstruct the phase information and therefore the 3D image using an iterative phase retrieval algorithm (IPRA). Recovering the full three-dimensional strain field requires repeating this measurement at three linearly independent Bragg peaks, as in Figure 2.

Recovering that phase is fundamentally ill-posed: a diffraction measurement gives only the magnitude of the object’s Fourier transform, and many different objects can share the same magnitude, so the phase cannot be inverted directly and must instead be recovered through iterative constraints [2, 3]. The Gerchberg–Saxton algorithm was the first practical solution, alternating between real and Fourier space and enforcing known constraints, such as positivity and a finite support, in each domain [2]. Fienup later reframed this alternating-projection idea into the error-reduction and hybrid input-output algorithms that underlie most modern IPRAs [4]. In practice, however, these projection algorithms are prone to stagnating in local minima or converging to spurious, non-physical solutions, particularly for symmetric or noisy samples, and even successful reconstructions typically yield only a smoothed estimate of the electron density rather than a fully atomistic structure. Overcoming these limitations, especially for imaging strain at atomic resolution, motivates the physics-informed approach described below.

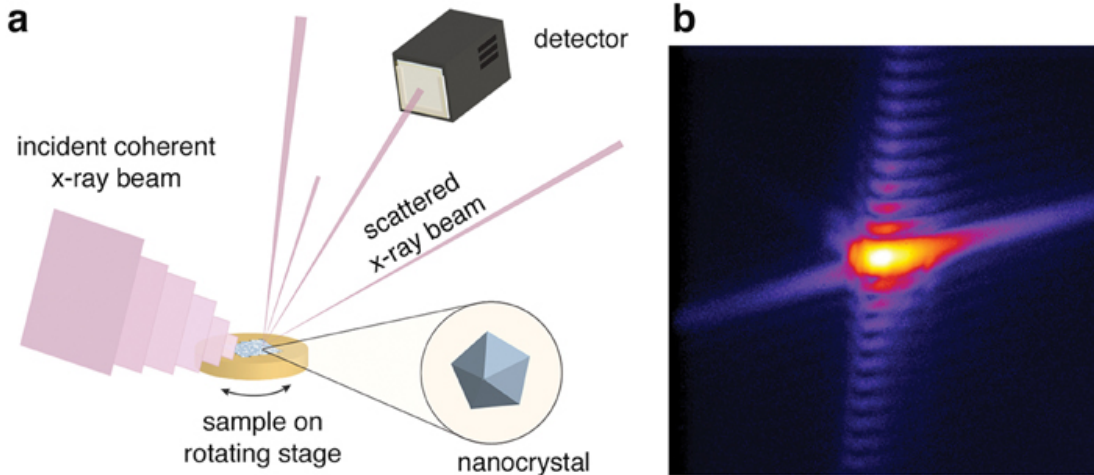


Figure 1: Coherent X-rays scatter from a nanocrystal; intensity is collected around a Bragg peak [5].

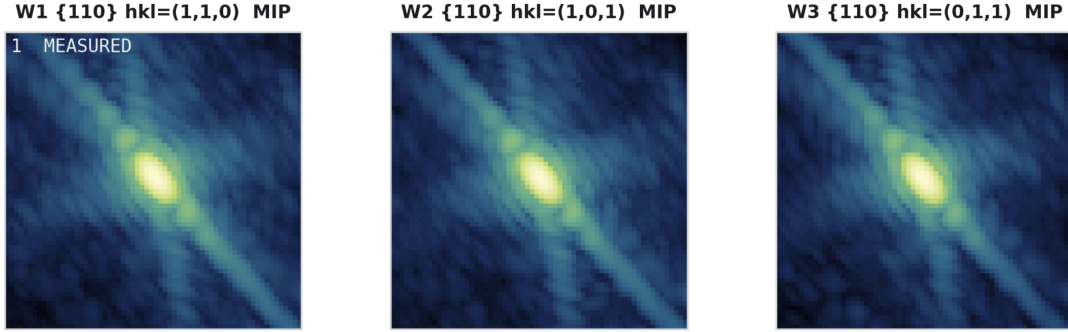


Figure 2: Measured diffraction from a tungsten nanocrystal at three $\{110\}$ peaks. Three linearly independent peaks are required to reconstruct the 3D strain field.

This report makes three contributions: a GPU- and NUFFT-accelerated reimplementaion of PRAMMol that makes reconstructions at the 5k–50k-atom scale computationally tractable; a characterization of how reconstruction convergence depends on grain size across a 144-run library; and the finding that radiation damage improves rather than degrades reconstruction fidelity, demonstrated on simulated tungsten grains relevant to both fission and fusion reactor first-wall materials.

2 Methods

The standard IPRA is Error Reduction (ER) [4] — an iterative projection method that is equivalent to gradient descent and prone to getting stuck in local minima [6]. Other techniques like hybrid input/output and shrinkwrap can help ER escape local minima, but they are limited in their effectiveness. Because of these limitations, standard IPRA only provide a low-resolution reconstruction of the grain structure, as shown in Figure 3.

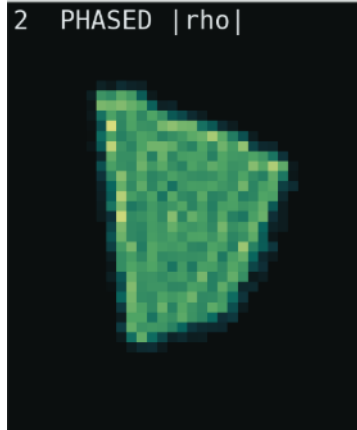


Figure 3: Low-resolution reconstruction.

These reconstructions provide valuable information but the utility is limited. By adding more information to constrain the search space we are able to resolve much finer detail in the structure.

The Approach: PRAMMol

Phase Retrieval with Atomic Modeling and Molecular Dynamics

PRAMMol is an in-house IPRA that embeds molecular dynamics (MD) directly in the solving loop, constraining every reconstruction to a physically possible atomic configuration [6]. Rather than reconstructing an electron density on a fixed voxel grid, as ER and its variants do, PRAMMol treats every atom as a free point in space and models the measured intensity ι_{hkl} at each point in reciprocal space as Poisson-distributed about $\lambda_{hkl} = |F_{hkl}|^2$, the squared magnitude of the structure factor

$$F_{hkl} = C \sum_j f_j \exp[2\pi i(hx_j + ky_j + lz_j)]. \quad (1)$$

F_{hkl} is the coherent sum, over every atom j of the N atoms in the object, at position (x_j, y_j, z_j) with atomic scattering factor f_j , of its contribution to the wave scattered toward reciprocal-space point (h, k, l) — effectively a Fourier transform of the atomic positions, and exactly a Discrete Fourier Transform when those positions and reciprocal-space points both fall on a regular grid. The measurement itself consists of N_{refl} such reciprocal-space points, referred to as reflections, each with its own measured intensity ι_{hkl} . It is the unknown atomic positions $(\vec{x}, \vec{y}, \vec{z})$, together with an overall scaling constant C , that PRAMMol treats as the parameters of that distribution [6]. Searching directly over atomic coordinates removes the pixel-resolution limit inherent to grid-based reconstructions and lets the algorithm draw on physical, materials-model information that ER has no mechanism to use.

Each outer iteration of PRAMMol proceeds in three stages [6], illustrated in Figure 4. First, a maximum-likelihood estimation (MLE) step updates the current atom positions and scaling constant by numerically maximizing the Poisson log-likelihood of the measured intensities, using F_{hkl} from Equation 1,

$$\ell(C, \vec{x}, \vec{y}, \vec{z} \mid \vec{v}) = \sum_{hkl} \iota_{hkl} \ln \lambda_{hkl}(C, \vec{x}, \vec{y}, \vec{z}) - \lambda_{hkl}(C, \vec{x}, \vec{y}, \vec{z}), \quad (2)$$

using only a small fraction of the diffraction pattern (about 0.5%) to keep this step computationally cheap. Second, new candidate atom positions are proposed — via Delaunay triangulation of the existing atoms, atomic force calculations, and random placement — and a cross-validation grid search decides which candidates to add and which existing atoms to remove, scoring each choice against the remaining 99.5% of the diffraction pattern held out as a validation set; this is the step that lets the total number of atoms grow, shrink, or stay fixed from one iteration to the next. Third, roughly half of the atoms, weighted toward those with the highest potential energy, are relaxed with a short LAMMPS MD run, pulling the structure back toward a physically realistic configuration before the cycle repeats. The loop terminates once cross-validation stops selecting new atoms.

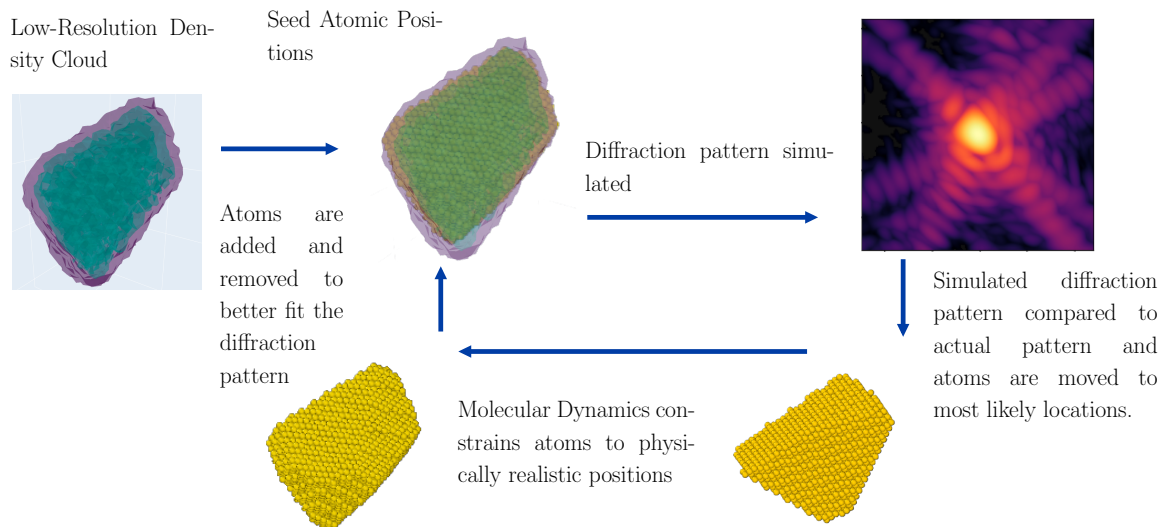


Figure 4: Visualization of the PRAMMol algorithm. Starting from a low-resolution density cloud, atomic positions are seeded within its envelope. A diffraction pattern is simulated from these positions and compared against the measured pattern; atoms are moved to the most likely locations. MD then constrains the atoms to physically realistic positions, and atoms are added and removed to better fit the diffraction pattern before the loop repeats.

Implementation: GPU Acceleration

Running PRAMMol at the $>10k$ atom scale needed for realistically sized grains required a substantial rewrite of the reference implementation, whose performance does not scale to those sizes. Four changes account for much of the improvement.

Gradient Computation

The reference recomputes the gradient of the log-likelihood (Equation 2) by creating a dense tensor of size $N_{\text{refl}} \times 3N \times 2$ on the GPU(device), transferring it to the CPU(host), and then reducing it on the host — hundreds of megabytes transferred between the device and host and reduced per line-search evaluation. The gradient can be factored algebraically into a sum over reflections of a per-atom term, so it can instead be computed with two GPU kernels: one reduces over atoms to form the global structure-factor sums, and a second, with one thread block per atom, reduces over reflections to form each atom’s three-component gradient. Keeping the reflection data resident on the device across the whole run, rather than re-uploading it every call, removes essentially all of the remaining transfer. Measured against the reference, this is $222\text{--}1017\times$ faster depending on GPU, and agrees with the reference to $10^{-13}\text{--}10^{-16}$ relative error.

NUFFT Transition

Evaluation of the structure factor by direct summation is an $O(N \cdot N_{\text{refl}})$ operation, and because N_{refl} must grow with the object size to avoid aliasing, this direct sum is effectively $O(N^2)$. This direct sum was replaced with a non-uniform fast Fourier transform (NUFFT): the forward structure factor becomes a type-1 NUFFT (atoms to a reciprocal-space grid) and the gradient a type-2 NUFFT (grid to atoms), reducing the cost to $O(N_{\text{refl}} \log(N_{\text{refl}}) + N)$. This was implemented first as a CPU backend and then as a GPU-resident backend that keeps the entire transform on the device, both validated to

within 10^{-9} – 10^{-11} relative error of the direct sum. Below roughly 1.5k–2k atoms the small reflection grid favors the direct sum, but above that the NUFFT backend pulls away sharply: $5.2\times$ at 5k atoms, $39.8\times$ at 20k, and $515\times$ at 100k, where a single structure-factor evaluation drops from 20 seconds to 39 milliseconds. This is the change that makes the largest grain sizes in this study computationally feasible at all, rather than simply faster.

Candidate Relaxation

Proposing a new atom position requires relaxing it to a local energy minimum under the interatomic potential, which the reference does by handing each candidate to a separate LAMMPS process. Most of that cost is not the relaxation itself: on a 131-atom test object, LAMMPS’s per-candidate setup — writing a data file, re-parsing the EAM potential table, and rebuilding the neighbor list and pair style from scratch — took 45.7 ms, while the conjugate-gradient minimization it existed to run took only 3.2 ms, so 93% of the wall time went into rebuilding an all but identical problem from scratch for every candidate. To eliminate the LAMMPS overhead, a self-contained custom GPU kernel that evaluates the embedded-atom-method (EAM) energy and forces directly was implemented using the same logic found within the LAMMPS MD engine. This allowed a whole batch of candidates to be relaxed on the device without leaving Julia or paying LAMMPS’s per-candidate setup cost. Validated against the LAMMPS implementation to within a few μeV , this candidate-relaxation step is roughly $13\times$ faster than the reference’s per-candidate LAMMPS process when run on the CPU, and roughly $15\times$ faster when run on the GPU.

Candidate Refinement

Once an add/remove candidate has been proposed, it is refined against the measured diffraction pattern; the reference does this with an LBFGS optimizer that is run independently for each candidate. This is expensive not because LBFGS is a poor choice of algorithm, but because of how the reference runs it: each candidate takes roughly ten outer LBFGS steps with a backtracking line search, 15–30 gradient evaluations in total, and each evaluation reuses the same general-purpose, full-object kernel used elsewhere in the code — resetting and re-summing the entire N_{ref} -length structure-factor arrays just to refine one atom’s three coordinates. Run across thousands of candidates, this becomes memory-bandwidth-bound: at 100k atoms it takes 150–173 seconds, the majority of the candidate stage, and adding CPU threads does not help, because every thread streams the same large arrays and saturates the same memory bus. Because each candidate is its own independent, sequential LBFGS trajectory, the reference’s own approach cannot be batched onto the GPU. But because each candidate has only three degrees of freedom, the underlying problem can be: this refinement is instead solved as a batched Gauss–Newton (GN) problem on the GPU, one thread block per candidate. Run to full convergence, this refiner overshoots the object’s diffraction field, which itself shifts slightly every outer iteration, increasing how much the reconstruction churns; the GN refiner therefore includes a damping cap that limits each step to a fraction of the local lattice spacing, recovering behavior close to the reference optimizer’s own deliberately gentle steps. On the same 30k-atom reconstruction, this alone cuts the refinement step from 28.0 to 0.81 seconds, roughly $35\times$.

Net Effect

Component	Comparison	Speedup
Structure-factor cost and gradient	GPU vs. reference (CPU)	222–1017×
Structure-factor evaluation, 100k atoms	GPU NUFFT vs. GPU direct sum	515×
Candidate relaxation (EAM)	Custom GPU kernel vs. per-candidate LAMMPS	~15×
Candidate refinement, 30k atoms	Batched GPU Gauss–Newton vs. per-candidate LBFGS	~35×

Table 1: Representative speedups from the GPU reimplementaion. The NUFFT transition additionally changes the scaling of structure-factor evaluation from $O(N^2)$ to $O(N \log N)$, which is what makes the largest grain sizes reachable at all.

By Amdahl’s law, a component’s effect on the net speedup is capped by the fraction of an iteration it occupies, not by its isolated speedup: the structure-factor kernels post the largest number in Table 1 (up to 1017×), but that math is only 1–2% of an iteration, while relaxation and refinement make up nearly all the rest and dominate the net effect instead, with refinement’s share growing from 9% to 31% between 10k and 30k atoms. A matched total at this study’s own grain sizes (5k, 25k, 50k) was not collected.

Application: Imaging Radiation Damage in Nuclear Reactor Materials

To test PRAMMol in an industrially applicable situation, it was used to image radiation damage in nuclear reactor materials. Fission and fusion reactors alike expose first-wall and structural materials like tungsten and steel to high-intensity radiation and thermal cycling, with the flux and damage rates expected in future fusion devices — particularly inertial fusion energy (IFE) concepts — pushing this exposure toward its most severe regime. Understanding the strain these conditions produce is vital to the design of both existing fission and next-generation fusion reactors, and ultimately to power generation more broadly: by understanding the strain, we can better predict the lifetime of these materials and design more resilient reactors, reducing capital expenditure and maintenance costs. To this end, a variety of tungsten grains were simulated across different sizes (5k, 25k, and 50k atoms) and then exposed to α radiation at various energies.

3 Results

Additional results and documentation from these and similar runs are available at <https://nelsbuhrley.github.io/PRAMMol/>.

How Size Affects Convergence

Two metrics are used throughout to score a reconstruction against the measured diffraction pattern. The R-factor, $R = \sum_r |A_r - n|F_r|| / \sum_r A_r$, is the total absolute deviation between the measured amplitude $A_r = \sqrt{I_r}$ and the scaled model amplitude $n|F_r|$, normalized by the total measured amplitude, where n is an overall fitted scale. The reduced χ^2 , $\chi_{\text{red}}^2 = \frac{1}{\text{dof}} \sum_r \frac{(\nu_r - \lambda_r)^2}{\lambda_r}$, is the Poisson-weighted squared residual per degree of freedom, with $\text{dof} = N_{\text{refl}} - 3N - 1$. Because shot noise alone keeps even the true atomic structure from fitting the data exactly, both metrics are also evaluated against each

grain’s own known ground-truth configuration to give a noise floor: the best score achievable given the measurement’s own statistics. A reconstruction whose χ^2_{red} is within order unity of this floor — written χ^2/floor — fits the data as well as the true structure does and cannot be meaningfully improved further; computing this floor requires the ground truth, so it is only available in this simulation study, not for a real measurement.

Figure 5 plots the R-factor and χ^2/floor against outer iteration for the 144 runs in the $\sim 5\text{k}$ -atom library, colored by radiation dose: *L0* (pristine, no radiation), *L1* (150 eV cascade), *L2* (400 eV cascade), *L3* (1 keV cascade), and *L3deep* (a deeper variant of the 1 keV cascade), in order of increasing radiation intensity. Converged runs often fall sharply within the first few tens of iterations and settle near the noise floor; unconverged runs plateau early and churn indefinitely, three to four orders of magnitude above the floor, often no matter how much longer they run. Despite this typical split between rapidly converging and churning, it is not uncommon to see a run churn for two or three hundred iterations and then rapidly converge to the correct answer. As grain size grows the chances of convergence decrease: roughly 70.5% of $\sim 5\text{k}$ -atom runs converged, versus 25.0% at $\sim 9\text{k}$ atoms, 6.6% at $\sim 25\text{k}$ atoms, and none of the 82 $\sim 50\text{k}$ -atom runs had converged by the time they were evaluated — though those largest runs were still visibly improving rather than stalled outright. For runs completely stuck in the churning state, the most reliable remedy is to run many reconstructions in parallel from slightly different initial conditions.

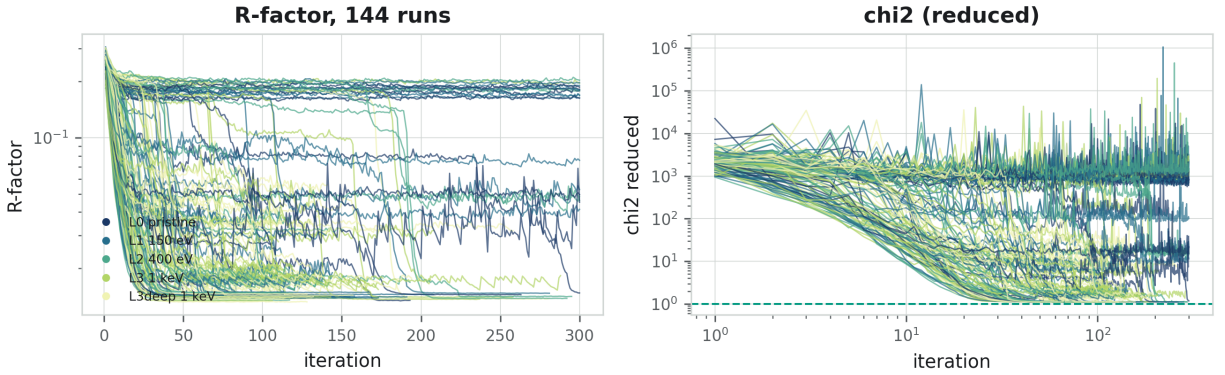


Figure 5: Convergence trajectories for the 144-run $\sim 5\text{k}$ -atom library, colored by radiation dose. Left: R-factor vs. outer iteration. Right: reduced χ^2 vs. outer iteration (log-log), with the noise floor as a dashed line.

How Radiation Damage Affects Convergence

Damage was expected to degrade reconstruction fidelity; instead it improved it. A single 22k-atom grain was damaged with a ladder of primary-knock-on-atom (PKA) cascade energies — 150, 400, and 800 eV — and each reconstruction was scored against that same grain’s own undamaged control. Vacancy count rose monotonically with cascade energy (23, 33, and 49 vacancies at 150, 400, and 800 eV respectively), but the undamaged control was the one that failed to converge (χ^2/floor 136.6, mean position error 0.6994 Å), while every damaged rung converged cleanly (χ^2/floor 2.5–4.3, mean position error 0.005–0.014 Å). The same pattern holds across the full library: Figure 6 plots final position error against atom-count error for all 144 runs, colored by dose, and both errors fall together as dose increases, from *L0* pristine (worst, upper right) to the 1 keV rungs (best, lower left). The likely explanation is that the vacancies and off-lattice displacements a cascade creates give the solver additional distinguishing structure to lock onto that a perfect lattice does not provide — an undamaged crystal is highly symmetric, and symmetry is exactly what makes phase retrieval ambiguous.

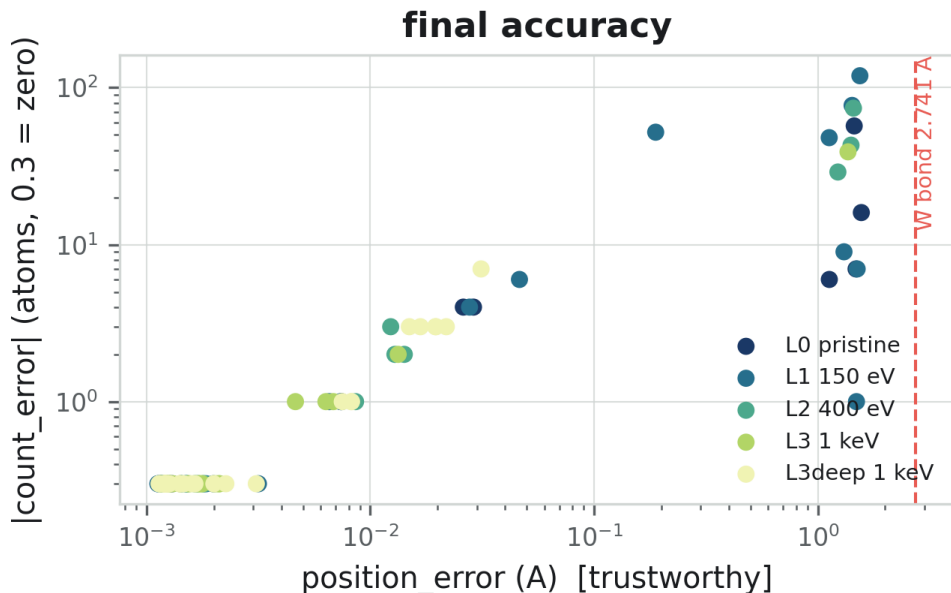


Figure 6: Final position error vs. atom-count error at the end of each run, for the same 144-run library, colored by radiation dose. Lower doses (dark) cluster at high error, upper right; higher doses (light) cluster at low error, lower left.

Example Reconstructions

PRAMMol reconstructed many of the simulated grains. Figure 7 makes the convergence signature described above concrete for two representative runs at different points across the size range. Figure 7a is a 5k-atom reconstruction that converged to the exact lattice structure, with per-atom position error confined to 0.00–0.08 Å across the grain; Figure 7b is a 25k-atom reconstruction that never fully converged, with position error reaching as high as 1.5 Å in places. The bottom two rows of each panel are the same atom-count and R-factor traces discussed above, made concrete for these two runs: the converged run’s count settles and its R-factor drops to the noise floor, while the unconverged run’s count and R-factor both continue to churn for the full length of the reconstruction.

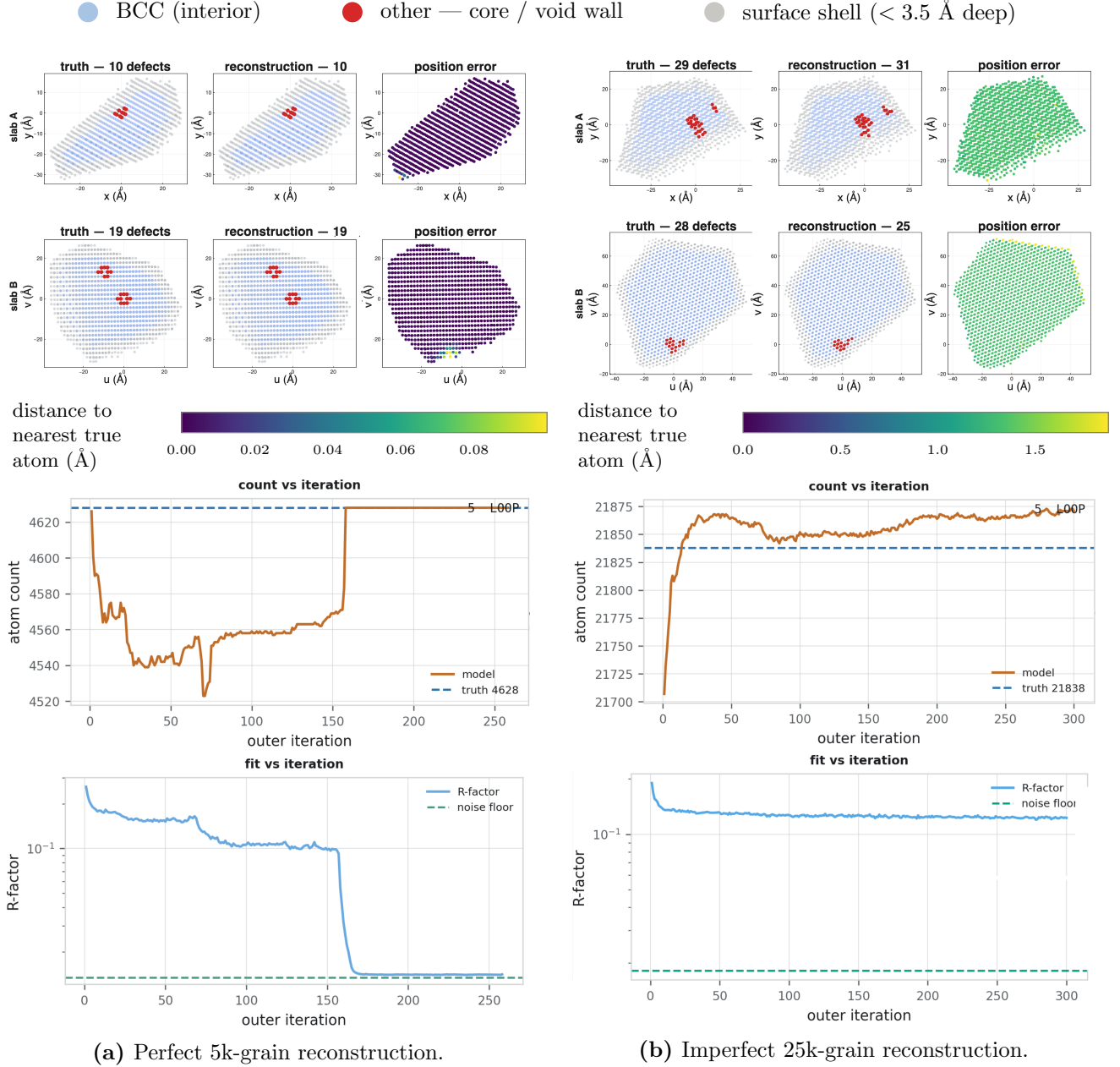


Figure 7: Two examples of reconstructions. Each panel shows, top to bottom: slices of the truth structure, reconstruction, and per-atom position error (colored by distance to the nearest true atom); the running atom count; and the R-factor, both plotted against outer iteration.

4 Conclusion

PRAMMol has been demonstrated as an IPRA that recovers atomic-resolution, three-dimensional structure directly from Bragg coherent diffraction data, using molecular dynamics as an explicit physical prior rather than a fixed-grid density estimate. The GPU and NUFFT rewrite described in Methods made reconstructions up to the 50k-atom scale used in this study computationally feasible, and convergence across the resulting archive is well characterized: most $\sim 5\text{k}$ -atom grains converge to the exact lattice structure, convergence rate falls off sharply as grain size grows, and — counterintu-

itively — radiation damage itself improved reconstruction fidelity rather than degrading it, most likely because the symmetry of an undamaged lattice is exactly what makes the phase problem ambiguous.

These results demonstrate PRAMMol’s potential to image the strain state of irradiated structural materials directly, at the atomic scale the damage itself occurs, in both existing fission reactors and next-generation fusion devices — with particular relevance to the more severe damage environments expected in inertial fusion energy (IFE) first-wall materials.

5 Future Work

Further Computational Optimization

Several stages of the outer iteration remain unaddressed by the GPU rewrite described in Methods. One primary example is the high-energy relaxation step: the third stage of each outer iteration, which relaxes existing atoms rather than newly proposed candidates, still dispatches to LAMMPS on the CPU every iteration. Unlike candidate relaxation, it is a single coupled many-body relaxation rather than thousands of independent three-degree-of-freedom problems, so the batched kernel used for candidates does not apply directly, and porting it would require a genuinely parallel many-body GPU molecular-dynamics implementation rather than a simple batching trick.

Beyond fixing what is already GPU-resident, sharding candidate relaxation and refinement across the multiple GPUs available on a single compute node, rather than running on one GPU at a time, is a comparatively low-risk way to add another constant-factor speedup. Another more speculative direction with a much higher ceiling is porting the structure-factor and candidate kernels to the Cerebras Software Language (CSL) to run on a wafer-scale engine, such as the NSF-funded Neocortex system at the Pittsburgh Supercomputing Center. PRAMMol’s candidate stage is exactly the kind of workload a wafer-scale architecture is built for: tens of thousands of small, structurally identical, independent problems, one per candidate, that on a GPU are ultimately limited by shared memory bandwidth, but that could instead be spread across a wafer’s hundreds of thousands of processing elements, each with its own local memory and a low-latency on-wafer mesh interconnect in place of a shared bus. This would require a substantial reimplementation — CSL is a low-level, explicitly scheduled language with none of Julia’s existing tooling — and has not been attempted or benchmarked here, but the memory-bandwidth ceiling identified in the refinement step is exactly the kind of bottleneck this architecture is designed to target.

Validation Against Measured Beamline Data

All of the results in this report come from simulated diffraction patterns generated from a known ‘ground-truth’ atomic configuration, which is what makes the exact atom-count and position-error metrics in the Results section possible in the first place — there is no ground truth to compare against for a real measurement. The next step is to run PRAMMol on diffraction data actually collected at a synchrotron coherent-diffraction beamline (34-ID-C @ APS), from a real nanocrystal grain. Real measurements introduce failure modes that simulation does not: detector background and shot noise beyond the idealized Poisson model already used, partial spatial and temporal coherence of the incident beam, sample drift over the hours a full three-peak measurement can take, and imperfect alignment of the three linearly independent Bragg peaks the reconstruction requires. We can do our best to simulate these challenges, but ultimately, demonstrating that PRAMMol converges on real data, and that the resulting strain field agrees with independent damage characterization of the same sample, would be the definitive test of whether the method is ready to image real nanostructure materials.

AI Disclosure

Generative AI was used throughout research, and in formatting and editing this report.

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